

Venkata Sai Kandipati

Java Backend Developer

Johannesburg, South Africa | +27 60 713 9095 | venkat6827@gmail.com

PROFESSIONAL SUMMARY

Java backend developer with 7+ years building production systems across three distinct domains: cloud-native payment processing on AWS serverless infrastructure, enterprise risk-analysis platforms, and industrial data integration (GE Historian, iFIX). Specialized in Java, Spring Boot, microservices architecture, RESTful API design, and AWS serverless cloud solutions (Lambda, Step Functions, SQS, SNS), translating complex business requirements into secure, maintainable, high-performing systems built on PostgreSQL, MySQL, and Oracle data layers.

Comfortable across the full software development lifecycle — from architecture and implementation through CI/CD deployment, production support, and performance tuning. Applies SOLID principles, clean-code practices, design patterns, and resilience engineering (Retry, Circuit Breaker, Rate Limiter) to keep platforms stable under load. Uses generative AI tools (ChatGPT, GitHub Copilot) as part of a modern, efficient development workflow.

CORE COMPETENCIES

Microservices Architecture & Design • RESTful API Design & Integration • Cloud-Native & Serverless Development • Distributed Systems • Resilience Engineering & Fault Tolerance • Secure Authentication & Access Control (JWT, RBAC) • Database Design & Query Optimization • SOLID Principles & Clean-Code Practices • Design Patterns & Object-Oriented Design • Test-Driven Development • CI/CD & Automated Delivery • Production Support & Root-Cause Analysis • Agile / Scrum Delivery • Performance Tuning at Scale

TECHNICAL SKILLS

Languages & Frameworks	: Java, SQL, Spring Boot, Spring Framework, Spring Data JPA, Spring Security, Hibernate, Java EE, Servlets, JSP, JDBC, JWT
Cloud (AWS)	: Lambda, EC2, Step Functions, SQS, SNS, Secrets Manager, CloudWatch
Databases	: PostgreSQL, MySQL, Oracle, Schema Design
Messaging & Testing Tools	: Apache Kafka, RabbitMQ, JUnit, Mockito, Postman
DevOps & Tooling	: Git, GitHub, GitHub Actions, Bitbucket, Jenkins, Maven, Docker, Kubernetes
API & Documentation	: Swagger / OpenAPI
Tools & Platforms	: IntelliJ IDEA, Eclipse, Jira, Confluence, Apache Tomcat, Linux, Bash, Windows
Productivity (Gen AI)	: ChatGPT, GitHub Copilot

PROFESSIONAL EXPERIENCE

Software Developer — TechValley PTY LTD

Jun 2024 – Present

Johannesburg, South Africa | Project: Transaction Workflow Orchestrator

A cloud-native, AWS serverless platform handling payment processing, transaction routing, receipting, reconciliation, and cash-flow management for enterprise financial clients, engineered for reliability and scale.

Key Responsibilities:

- **Architected and built** Spring Boot microservices for payment processing, transaction routing, reconciliation, and receipting, using Spring Data JPA and Hibernate against a PostgreSQL data layer to keep transaction records consistent and auditable.
- **Owned the integration** of payment gateways, DebiCheck, and AVS verification services through secure REST APIs, enabling automated end-to-end payment validation.
- **Designed and implemented** AWS Step Functions and Lambda-based orchestration to automate multi-step transaction workflows, replacing manual processing with a resilient, event-driven pipeline.
- **Engineered** Retry, Circuit Breaker, and Rate Limiter resilience patterns at the service layer to protect the platform against downstream payment-provider failures.
- **Led root-cause investigation** of production incidents using Amazon CloudWatch log analysis, driving durable fixes that reduced recurrence rather than one-off patches.
- **Optimized** SQL queries and transaction-handling logic in PostgreSQL to sustain performance as transaction volumes scaled, partnering with the DBA team on schema decisions.
- **Established** secure configuration practices using AWS Secrets Manager, removing credentials from source control and simplifying rotation across environments.
- **Documented and published** RESTful APIs with Swagger/OpenAPI, enabling internal and partner teams to self-serve integration.
- **Worked directly** with Business Analysts and QA to translate payment-processing requirements into technical designs, keeping delivery aligned through sprint planning and reviews.

Technical Environment: Java, Spring Boot, Spring Data JPA, Hibernate, REST APIs, PostgreSQL, AWS Lambda, AWS Step Functions, AWS Secrets Manager, CloudWatch, Docker, GitHub Actions, Swagger, JUnit, Mockito, Git.

Software Developer — Genpact PTY LTD

Apr 2022 – May 2024

Hyderabad, India | Project: Risk Manager

An enterprise software risk-analysis platform used by global development teams to identify application vulnerabilities through static and dynamic code analysis, automating risk assessment, remediation tracking, and compliance reporting.

Key Responsibilities:

- **Developed** backend components in Java, Spring Boot, and Spring Data JPA to support the platform's risk-assessment features against a MySQL data layer.
- **Built** reusable business services and utility components following SOLID principles and object-oriented design, reducing duplication and speeding up delivery of related features.

- **Implemented** authentication and authorization using Spring Security and JWT, applying role-based access control (RBAC) to protect vulnerability and compliance data.
- **Wrote and maintained** unit and integration tests with JUnit and Mockito, improving code quality and catching regressions before QA.
- **Tested and validated** APIs throughout development using Postman, helping keep API contracts consistent across services.
- **Supported** CI/CD pipeline delivery with Git, Maven, and Jenkins, contributing to repeatable builds and releases.
- **Worked alongside** QA in sprint planning and code reviews, resolving production defects tied to risk-assessment accuracy and compliance-reporting logic.

Technical Environment: Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, MySQL, REST APIs, JUnit, Mockito, Postman, Maven, Jenkins, Git.

Junior Software Developer — Genpact PTY LTD

Mar 2019 – Mar 2022

Hyderabad, India | Project: OpsHub

A centralized operations platform integrating GE Digital's Historian and iFIX products to collect, process, and visualize industrial data, delivering real-time operational insight to manufacturing teams through monitoring dashboards and reporting.

Key Responsibilities:

- **Gained hands-on exposure to** Java, Spring Boot, JPA, and Hibernate while assisting with backend features for industrial data processing and reporting, deployed on Apache Tomcat.
- **Assisted with** integrating GE Historian and iFIX systems for real-time data collection, working closely with senior engineers to learn the underlying data models.
- **Supported** SQL query writing and tuning against MySQL for data retrieval and reporting, helping feed the monitoring dashboards used by manufacturing teams.
- **Communicated** with senior engineers and QA during deployment, environment validation, and defect resolution, contributing to day-to-day platform stability.
- **Tested** API endpoints with Postman under senior guidance to confirm service reliability ahead of release, maintaining a record of test cases.
- **Built early exposure to** enterprise SDLC and Agile delivery through the OpsHub project, learning how industrial data systems are tested, deployed, and supported in production.

Technical Environment: Java, Spring Boot, JPA, Hibernate, MySQL, REST APIs, Apache Tomcat, Maven, Git, Postman.

ACADEMICS

Acharya Nagarjuna University - India
BCom (Computer Applications)